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CardioDiverse: A Balanced ECG Dataset for Improved Association of Heart Diseases with Relevant Leads

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***Abstract*—This study introduces CardioDiverse, a new dataset that has been constructed to address the scarcity of balanced data in existing electrocardiogram (ECG) datasets. CardioDiverse is a combination of multiple datasets integrated to provide rich information to support the association between the different ECG leads and CVD subclasses. The dataset is available to the public through GitHub. Our findings illustrate that particular leads provide clear manifestation to certain CVD subclasses. For instance, leads II, AVF, V3, V4, V5, and V6 were predominantly associated with ST-T wave abnormality, whereas leads II, III, AVF, V1, V5, and V6 were more affiliated with hypertrophy. Furthermore, by extending the analysis to include specific subclasses within the PTB-XL dataset, our study shows promising results in categorizing these subclasses using single-lead ECGs. CardioDiverse is an invaluable resource for researchers, as it provides a more balanced dataset and includes essential data analysis tools within the repository. The data and insights obtained through this study pave the way for future research to investigate the use of these leads for the early detection and diagnosis of cardiovascular diseases.**

***Index Terms*—Lead grouping, ECG, cardiovascular diseases, dataset, ECG lead correlation, ST-T wave abnormality, single-lead ECG, heart diagnosis, early detection.**

I. Introduction

Cardiovascular diseases (CVDs) represent a global health challenge, accounting for almost 17.9 million deaths worldwide in 2019 [1]. Addressing this growing threat lies in early detection and precise diagnosis, which improves patient outcomes and mitigates the burden on healthcare infrastructures. Electrocardiogram (ECG) emerges as a key in this endeavor, monitoring heart electrical activities and providing essential information on its functioning [2]. Thus, continuous cardiac monitoring becomes an urgent requirement to quickly identify abnormalities and facilitate immediate medical attention [3]. However, seamless and efficient real-time monitoring remains challenging due to either the financial costs associated with monitoring equipment or the durability of battery-powered ones, both essential for the efficacy of ongoing monitoring [3]. Furthermore, transmission latency, or the delay between data capture by the monitoring device and its receipt by healthcare professionals is another significant factor that affects the effectiveness of real-time surveillance [3] [4]. To overcome these obstacles, recent trends are geared towards reducing the number of leads employed in ECG monitoring. However, conventional ECG monitoring relies on 12 leads, each showing a unique perspective on heart electrical activities. Using fewer leads can reduce both the complexity and costs associated with monitoring setup and alleviate data transmission requirements [3] [4]. One proposition gaining traction is lead fusion, which entails merging several leads into a combined signal, thereby retaining diagnostic accuracy while cutting down the number of leads [5] [6]. However, lead fusion presents its own set of challenges, including optimizing lead configurations for a wide range of cardiac conditions. The limited availability of diverse datasets, which adequately represent various cardiac diseases and their relationships with different leads, further escalates these challenges [5]

The present study contributes by building a new data set that combines existing widely used ECG databases to ensure a heterogeneous and balanced representation of various cardiac problems. The study also identifies relationships between particular leads and diseases using comprehensive 12-lead ECG data. The data set assembled in this study paves the way for future exploration of the complications that arise in lead fusion, especially in classifying ECG signals instead of using the entire array of 12 leads.

The remainder of this paper is organized as follows. Section II reviews the relevant literature on existing ECG databases and outlines their limitations. In Section III, we analyze popular ECG databases and identify their strengths and weaknesses. Section IV introduces our new database, CardioDiverse, detailing the methodology and criteria for its creation. In Section V, we present our analysis and findings using CardioDiverse, which include diagnostic accuracy for various cardiac conditions using different lead configurations. Finally, in Section VI, we summarize the key contributions of this paper and draw future research directions.

II. Background and Related Work

ECG is a widely used diagnostic tool in cardiology to detect abnormalities in the electrical activity of the heart. Traditionally, ECG monitoring uses 12 leads to provide a comprehensive view of the electrical activity of the heart. However, the use of fewer leads can help reduce the cost and complexity of monitoring devices and minimize the time it takes to transfer data [3], [4]. Consequently, there has been growing interest in reducing the number of leads used for ECG monitoring. This has led to the development of several techniques, including lead grouping, lead selection, and deep learning algorithms, to improve the accuracy of ECG classification while reducing the number of leads used. This paper provides an overview of these techniques and their effectiveness in reducing the number of leads required for ECG monitoring. In addition, we highlight the challenges associated with lead grouping and selection methods and discuss the potential of deep learning algorithms for ECG classification.

A. Lead grouping

One approach to reducing the number of leads is lead grouping, which involves combining several leads to create a composite signal. This approach has been shown to provide similar diagnostic accuracy compared to using all 12 leads [5], [6]. However, lead grouping also presents challenges, such as difficulty determining the optimal lead grouping configuration and the potential loss of diagnostic information [5].

Several studies have investigated the performance of different lead grouping methods for ECG classification. For example, Li et al. [7] proposed a lead grouping method based on principal component analysis (PCA) to reduce the number of leads required for detecting atrial fibrillation. The authors showed that their method achieved an accuracy of 93.2% using only four leads compared to 95.4% using all 12 leads. Similarly, Chouvarda et al. [8] developed a feature extraction method for ECG signals using four-lead recordings, achieving an accuracy of 96.63% for diagnosing coronary artery disease.

Ponard et al. [9] proposed a novel approach to determine the optimal lead grouping configuration for ECG classification using machine learning algorithms. The authors evaluated their method on the PTB-XL dataset and achieved a classification accuracy of 92.8% using only three leads compared to 94.1% using all 12 leads. In a similar study, Zhou et al. [10] investigated the performance of different lead grouping methods to detect cardiac arrhythmias. The authors compared several grouping configurations and found that a combination of leads II, V2, and V5 achieved the highest accuracy of 97. 7%.

B. Lead selection

Lead selection is a crucial step in ECG classification and analysis. Various studies have investigated the effectiveness of different lead combinations and selection methods for accurate ECG classification. For instance, Muezzinoglu et al. [11] proposed a lead selection method based on mutual information and achieved an accuracy of 95.3% for detecting atrial fibrillation using only four leads. In another study, Zhang et al. [12] used a similar approach to identify the most informative leads for the diagnosis of myocardial infarction and achieved an accuracy of 91.2% using only three leads. Jiang et al. [13] proposed a method for lead selection using deep learning techniques to identify the most informative leads to detect arrhythmias. The study achieved an accuracy of 92.3% using only four leads. Similarly, Zhang et al. [14] used deep learning techniques for lead selection to detect myocardial infarction and achieved an accuracy of 95.6% using only three leads.

Another approach for lead selection is based on optimizing a cost function that incorporates both the diagnostic accuracy and the number of leads used. Such an approach was proposed by Khamis et al. [15] for detecting atrial fibrillation, where the optimal lead combination was identified based on a genetic algorithm. The study achieved an accuracy of 96.9% using only three leads.

In addition to the above studies, various other methods have been proposed for lead selection, including entropy-based approaches [16], correlation-based approaches [17], and principal component analysis-based approaches [18]. These studies have demonstrated the importance of lead selection in improving the accuracy of ECG classification while reducing the number of leads used.

C. Deep learning algorithms

In addition to lead grouping and lead selection methods, deep learning algorithms have also been applied to ECG classification tasks. Hannun et al. [18] developed a deep neural network model to detect 14 different ECG abnormalities using a single lead, achieving an accuracy of 90 2%. Similarly, Attia et al. [19] developed a deep learning algorithm for identifying atrial fibrillation using single-lead ECG recordings, achieving an accuracy of 83.9%. Other studies have investigated the use of convolutional neural networks (CNNs) for ECG classification. For example, Rajpurkar et al. [20] developed a 34-layer CNN model that achieved an accuracy of 95 6% to diagnose 12 different ECG abnormalities using all 12 leads. Guo et al. [21] proposed a CNN-based method for detecting atrial fibrillation using only one lead, achieving an accuracy of 98.5%.

While there have been several large publicly available datasets for electrocardiography (ECG) analysis, the comprehensive study of lead grouping has been limited due to the need for balance in data for a large number of classes.

III. Electrocardiogram Databases

ECG databases play a critical role in developing and testing algorithms for ECG analysis. These databases contain large amounts of ECG data recorded from individuals with various cardiovascular conditions, making them invaluable resources for research and clinical applications.

A. 12-lead ECG datasets

- **ECG Arrhythmia Database [22]:** The ECG Arrhythmia Database is a rich repository of ECG recordings, hosting 48 half-hour excerpts originating from 47 individuals. It's a treasure trove for researchers as it contains standard rhythms and is annotated for various arrhythmias and normal and abnormal beats. The granular annotation can serve as a potent tool for developing, training and validating arrhythmia detection algorithms. Furthermore, with half-hour data for each recording, researchers can observe and analyze longer cardiac activity patterns.
- **ECGDataDenoised [23]:** The ECGDataDenoised dataset entails 90 ECG recordings that have been meticulously

denoised. This offers a pristine quality of signals which is crucial for various applications including robust feature extraction. High-fidelity data is vital to enhance the performance of ECG analysis algorithms, particularly in environments where subtle variations in the signal could be diagnostically significant.

- **Georgia 12-Lead ECG Challenge Database [24]:** With 1000 ECG recordings encompassing many cardiac conditions and healthy individuals, the Georgia 12-Lead ECG Challenge Database is instrumental in algorithm evaluation. The diversity in cardiac conditions coupled with a large volume of data helps in the comprehensive assessment of algorithms under various scenarios and conditions, making it invaluable for refining the robustness and accuracy of ECG analysis systems.
- **CODE-test [25]:** The CODE test data set comprises 1000 recordings of 280 patients, annotated for different abnormalities, as well as normal ECG signals. This data set serves as an extensive base for researchers to investigate the identification of common and rare cardiac abnormalities. The varied patient pool aids in creating algorithms that are sensitive to patient-specific variations.
- **Annotated 12-Lead ECG dataset [26]:** This dataset is made up of 150 ECG recordings from 50 patients with various cardiac conditions. Each recording in this dataset is annotated for various cardiac abnormalities, providing researchers with a concentrated dataset for training and evaluating algorithms, particularly in the context of multi-class classification.
- **KURIAS-ECG [27]:** KURIAS-ECG provides 500 ECG recordings from 500 different patients. It distinguishes itself by incorporating standardized diagnosis codes in the annotations, which is significant for the development of diagnostic systems that can seamlessly integrate with healthcare information systems.
- **4th China Physiological Signal Challenge 2021 and CPSC2020 [28] [29]:** These datasets include ECG recordings from patients with cardiac conditions such as heart failure, ventricular arrhythmias, myocardial infarction, and atrial fibrillation. Including a range of severe cardiac conditions makes these datasets quintessential for developing critical care monitoring systems.
- **PTB-XL [30]:** The PTB-XL database is the status quo for ECG analysis and by far the most commonly used dataset by researchers and clinicians. The database contains more than 21,000 ECG recordings from over 5,000 patients with cardiovascular conditions, including myocardial infarction, congestive heart failure, and arrhythmias. The recordings are collected using standard 12-lead ECG machines and sampled at a frequency of 1,000 Hz, ensuring that high-quality recordings are suitable for the development and testing of algorithms for ECG analysis. The database is accompanied by expert cardiologist annotations, providing accurate diagnoses and annotations for cardiac abnormalities, further improving its usability. The PTB-XL database also contains demographic and clinical information for each patient, which can help researchers explore the relationship between patient characteristics and ECG findings, leading to the development of personalized diagnostic and treatment strategies.
- **PhysioNet Challenge 2020 [31]:**

B. What is missing?

Numerous studies have used various databases extensively, especially to develop and evaluate machine learning algorithms aimed at the classification of ECG. Among these, the PTB-XL database is one of the most used data sets. Researchers have used this data set to craft innovative methodologies for ECG classification, including detection of arrhythmias, analysis of ST segments, and identification of ischemia. For example, a series of recent research has deployed the PTB-XL database to scrutinize the configurations of lead grouping in ECG classification, managing to attain high accuracy with a limited number of leads. However, it should be noted that although PTB-XL is the largest publicly available ECG data set now, it encompasses only a subset of potential cardiac conditions. This renders limitations to the classification performance of developed algorithms. Consequently, there is a pressing need to expand the database by incorporating a more diverse array of cardiac diseases. This enhancement is essential for elevating the accuracy of ECG classification and gaining a better understanding of how the dataset can influence the outcomes of lead grouping. In this context, PTB-XL will serve as a foundational example to which we will compare CardioDiverse.

Class imbalance is regarded as one of the inherent limitations in existing datasets that can hurt the lead grouping process in ECG analysis. When a dataset exhibits class imbalance, it means that some categories of heart conditions are substantially underrepresented compared to others. This disparity in representation can cause algorithms to become biased towards the classes with higher representation, which results in the lead grouping process prioritizing leads that are more effective in identifying the overrepresented conditions. Consequently, the ability of the algorithms to accurately detect and classify the underrepresented heart conditions is diminished.

I shows the class imbalance in the PTB-XL dataset for its five superclasses: Normal, Myocardial Infarction (MI), ST-T wave abnormality or T wave inversions (STTC), Conduction disturbance (CD), and Hypertrophy (HYP). The table presents the total number of samples for each superclass, the number of positive and negative samples, and the corresponding imbalance ratio. It is clear from the I that the normal superclass has a relatively balanced ratio of positive to negative samples, whereas the other four superclasses have a significant class imbalance. For example, the MI superclass has an imbalance ratio of 1:3.39, indicating over three negative samples for every positive sample of MI. Similarly, the STTC superclass has an imbalance ratio of 1:3.21, and the CD superclass has an imbalance ratio of 1:3.86. The HYP superclass, on the other hand, has a slight class imbalance with an imbalance ratio of 1:1.26.

The imbalance ratio is calculated as the ratio of the number of negative samples to the number of positive samples for a given class. The positive and negative samples are defined based on the labels of the samples. A sample is considered

TABLE I: Imbalance Of The PTB-XL Dataset By Superclasses.

Superclass	Total	Positive	Negative	Imbalance Ratio
Normal	20511	8806	11705	1.33
MI (Myocardial Infarction)	5409	1230	4179	3.39
STTC (ST-T Wave Abnormality Or T Wave Inversions)	7658	1819	5839	3.21
CD (Conduction Disturbance)	1137	234	903	3.86
HYP (Hypertrophy)	2928	1293	1635	1.26
Total	36943	12382	24561	1.98

positive if it belongs to the target class and negative otherwise. For example, in the MI superclass, a sample is considered positive if it corresponds to a patient with myocardial infarction and negative if it corresponds to a patient without myocardial infarction. Generally, a balanced ratio close to one is desirable, indicating an equal number of positive and negative samples. The last row of Table I shows the total number of samples, positive samples, negative samples, and the imbalance ratio between all categories of superclasses. The difference between the total and total sample in the dataset is due to multiple labels for the sample in the dataset.

C. CardioDiverse

CardioDiverse combines several datasets that contain 29 distinct heart disorders, resulting in a more standardized dataset that includes 109 varied cardiac conditions. This extensive dataset provides a valuable resource for researchers and healthcare professionals interested in developing and testing algorithms for automatic ECG analysis and diagnosis. Including additional diseases in the CardioDiverse dataset will provide a more diverse range of ECG recordings for developing and testing arrhythmia detection, diagnosis, and classification algorithms. Researchers and clinicians can now explore the relationship between patient characteristics and ECG findings for cardiovascular conditions. This information can be used to develop personalized diagnostic and treatment strategies for patients with various cardiovascular conditions, leading to improved patient outcomes.

IV. The creation of CardioDiverse

An essential aspect of combining available ECG datasets into CardioDiverse was harmonizing the representation of various cardiac conditions. We matched disease subtypes across all datasets and unified their nomenclature and classification. Moreover, we balanced the dataset representation to avoid biases towards any specific condition. The CardioDiverse dataset contains over 200,000 ECG recordings from over 6,000 patients. It includes a wide spectrum of cardiac conditions, from common arrhythmias to severe cardiac ailments such as myocardial infarction and heart failure. It also comprises normal ECG recordings from healthy individuals. It also encompasses patient demographic and clinical information to support personalized analysis and predictions.

After building the CardioDiverse dataset, a series of signal pre-processing steps were executed to standardize all recordings. The objective was to ensure uniformity in the sampling frequency and duration in all ECG traces. Specifically, each recording was standardized to a length of 10 seconds and resampled at a frequency of 1000 Hz to be consistent with the format used by the PTB-XL dataset. In addition, disease subtypes across both datasets were thoroughly matched to ensure that each cardiac condition was represented uniformly. This curates a more complete, balanced, and comprehensive ECG dataset for advanced analysis.

Subsequently, we constructed lead grouping lookup tables for both the CardioDiverse and the PTB-XL datasets to compare their performance. The primary focus is to investigate how balancing the classes and the addition of new classes influence the results of lead grouping. This step is crucial to gain insights into the impact of dataset diversity and class representation on the learning process and model performance.

A. Performance utility metric

We tested several ECG algorithms to evaluate the performance of CardioDiverse as a benchmark using the class imbalance (U) metric, which is defined as:

$$U = \frac{2 \cdot (1-\beta^2) \cdot TP}{2 \cdot (1-\beta^2) \cdot TP + \beta^2 \cdot FN + FP} \tag{1}$$

where TP is the number of true positive predictions, FP is the number of false positive predictions, and FN is the number of false negative predictions. The parameter β controls the trade-off between precision and recall, with higher values of β emphasizing recall over precision. We set β to 0.5 to give equal weight to precision and recall.

To account for class imbalance, we use a weighted version of the utility metric that assigns a higher weight to the minority class, defined as:

$$U_w = \frac{2 \cdot (1-\beta^2) \cdot w_p \cdot TP}{2 \cdot (1-\beta^2) \cdot w_p \cdot TP + \beta^2 \cdot w_n \cdot FN + w_n \cdot FP} \tag{2}$$

where w_p and w_n are the weights of the positive and negative classes, respectively. We set w_p as the inverse of the proportion of positive samples in the training set and w_n as the inverse of the negative samples in the training set.

The utility metric provides a comprehensive evaluation of the performance of ECG algorithms on the CardioDiverse benchmark, considering both accuracy and class imbalance. It can be used to compare the performance of different algorithms and optimize the hyperparameters of machine learning models.

CardioDiverse is publicly available at https://github.com/ahmadammar972/12_lead_ECG_Dataset.git. The GitHub repository contains two folders for each dataset, CardioDiverse and PTBXL. Each folder contains 13 files for each lead (I, II, III, AVF, AVL, AVR, V1, V2, V3, V4, V5, V6) and the corresponding label. The data is sorted into the same indexes, and no metadata is provided, as it was created to provide a general idea of the relationship between leads and diseases.

The GitHub repository includes two starter codes to customize the merge style and to read the data. CardioDiverse provides a more balanced and comprehensive resource for researchers to investigate the correlation between leads and multiple diseases.

V. Experiments

A. Evaluating class balance

Analyzing the data balance within the CardioDiverse dataset is of paramount importance, particularly when it is deployed to consolidate crucial data in ECG analysis, and more specifically in creating groups of leads based on their efficacy in disease classification within each major category. An imbalance, characterized by a disproportionately large number of instances of certain sub-diseases in contrast to a scant number of others, can skew the results. This imbalance might cause the leads reflecting the predominant category to be biased toward the group with the highest representation. To ensure that the selected leads are optimally representative of each of the superclasses, it is essential to maintain a well-distributed and balanced assortment within each group.

The balance was examined for five superclasses: Normal, MI (Myocardial Infarction), STTC (ST-T wave abnormality or T wave inversions), CD (Conduction disturbance), and HYP (Hypertrophy) Table II shows the total number of samples, positive samples, negative samples, and imbalance ratio for each superclass. An imbalance ratio close to 1 indicates a balanced subclass, while a value much larger than 1 indicates an imbalanced subclass. According to Table II, the total imbalance ratio for the combined dataset is 0.41, indicating a relatively balanced dataset.

TABLE II: Balance of The CardioDiverse Dataset by Superclasses.

Superclass	Total	Positive	Negative	Imbalance Ratio
Normal	54365	28641	25724	1:1.11
MI (Myocardial Infarction)	8627	1844	6783	1:3.68
STTC (ST-T Wave Abnormality Or T Wave Inversions)	22952	5747	17205	1:2.99
CD (Conduction Disturbance)	3296	830	2466	1:2.97
HYP (Hypertrophy)	7048	3098	3950	1:1.28
Total	97188	40660	56528	1:1.39

The superclass with the highest imbalance ratio is MI, with a value of 2.717. This indicates a severe imbalance between positive and negative samples in this superclass. On the other hand, the superclass with the lowest imbalance ratio is HYP, with a value of 0.136, indicating a more balanced distribution of positive and negative samples.

As shown in Table III, CardioDiverse significantly improves the balance of several superclasses, such as MI, STTC, CD, and HYP. For example, the imbalance ratio of the MI superclass decreased from 9.17 in the PTB-XL dataset to 2.72 in the combined dataset, indicating a much more balanced distribution of positive and negative samples. Similarly, the imbalance ratio of the CD superclass decreased from 1.27 in the PTB-XL dataset to 0.50 in the combined dataset. Moreover, CardioDiverse introduces new positive samples for the Normal and HYP superclasses, which were not present in the PTB-XL dataset.

B. CardioDiverse versus PTB-XL

In this study, we generated a correlation table between heart diseases and relevant ECG leads to improve the accuracy and efficiency of diagnosing cardiovascular diseases. In this experiment, we used both the PTB-XL and the CardioDiverse dataset to build a correlation with the disease-lead correlation table and compared their accuracy when used for classification. We performed the analysis separately for each of the four superclasses in the PTBXL dataset: MI, HYP, STTC, and CD. A total of 4,095 possible combinations of the 12 leads were generated and classified for each superclass using the xresnet18 classification algorithm. The resulting association table included the group of leads and their corresponding accuracy, which was used to calculate the correlation between each lead and the disease.

The analysis then focused on identifying the best group of leads that represent the superclass by analyzing the correlation between the leads. This involved analyzing the accuracy for each group of leads and determining which group yields the best overall performance. The resulting correlation table provided information on the most effective leads for each superclass, allowing a more targeted and accurate analysis of the ECG data.

C. Case study: CardioDivase covers more diseases

In this experiment, we demonstrate that CardioDiverse covers more spectrum of heart conditions than PTB-XL. We used the thesis of Mousa et al. [34] since they showed that only six leads are required to efficiently classify heart conditions without significant loss in accuracy compared to 12 leads. We use the superclass *Conduction Disturbance (CD)* heart condition as an example, but all five superclasses yield similar results. Thus, we first identify the six main relevant leads that contribute to the detection of CD by analyzing all possible combinations of leads in both PTB-XL and CardioDiverse. Our analysis shows that the PTB-XL dataset includes 11 subclasses for the CD superclass. Three of these subclasses contain fewer than 100 samples, thus we have to remove them from the analysis to avoid underfitting. CardioDiverse, on the other hand, mitigates this data imbalance and enables the inclusion of all 11 subclasses, thus covering the missing three from PTB-Xl. This is a significant boost in ECG training and detection, resulting in about 30% more coverage for CD subclasses.

Subsequently, we investigate whether the inclusion of these subclasses has any discernible impact on the results obtained from our lead combination analysis. To this end, this experiment performs binary classification (i.e., distinguishing only between normal and CD conditions), utilizing all possible combinations of the six leads from the set of 12 as outlined by Mousa et al. [34].

Table III compares lead sequence and their associated accuracy for all patients with CD heart conditions.

TABLE III: The Most Correlated Leads With Conduction Disturbance (CD) in PTB-XL and CardioDiverse.

Dataset	**Top six leads relevant to CD ordered by their contribution**	**F1-score**
PTB-XL	II, III, AVF, AVR, V1, V2	94.79
CardioDiverse	II, V5, III, AVF, V1, V2	93.80

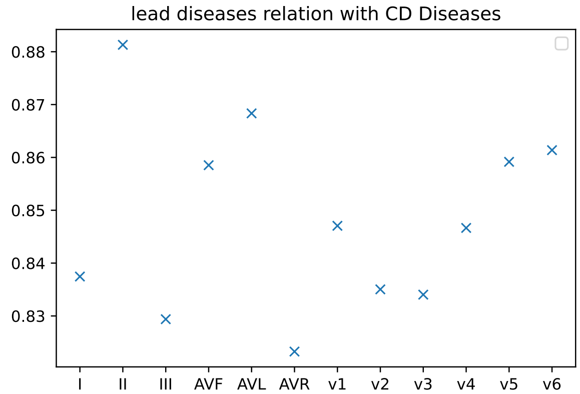

Fig. 1: The Individual F1-scores for Each Lead.

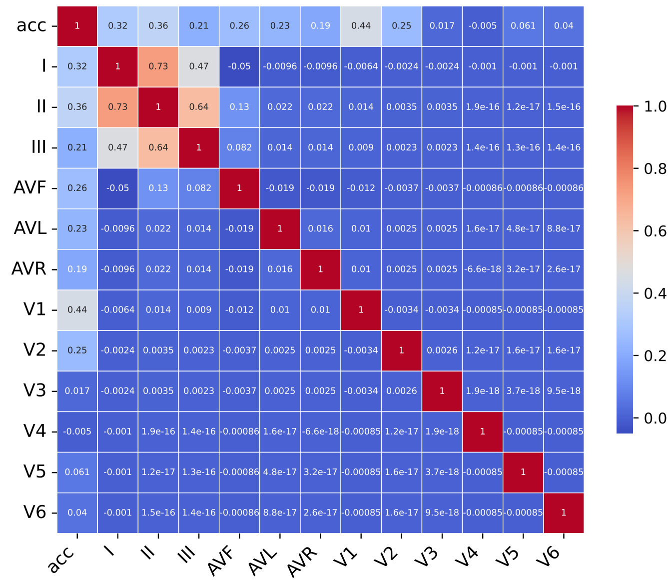

Fig. 2: Disease-Lead Correlation Matrix For the Conduction Disturbance superclass.

Table III indicates that CardioDiverse yields a comparable F1 score to that of PTB-XL, but offers a better balanced distribution of data and diversity within subclasses. Figure 1 shows the individual F1 scores for each lead, showing that lead II, AVL, V5, and V6 are the most effective leads for the CD superclass.

To further explore the relationship between leads and CD conditions, we generated a correlation matrix using all possible combinations of leads and accuracy measures. Figure 2 presents the resulting disease-lead correlation matrix for CD. The matrix depicts the correlation between the disease and each lead, along with the strength of the correlation. This information can help identify the most relevant leads for detecting CD conditions and improve diagnostic accuracy. According to the correlation matrix, the top leads are Lead II, Lead III, Lead AVF, Lead V1, Lead V2, and Lead V5.

1- Lead II is highly correlated with Leads I and III and several precordial leads (V1, V2, V5, and V6), making it a strong candidate for representing Conduction disturbance.

2- Lead III has a high correlation with Lead II and Lead AVF, and a moderate correlation with Lead I and several precordial leads (V1, V2, V3, and V6), making it another good candidate for representing Conduction disturbance.

3- Lead AVF has a high correlation with Lead III and a moderate correlation with Lead II and Lead AVL.

CD also has a low to moderate correlation with several precordial leads (V1, V2, V3, and V4), making it a possible candidate to represent the disease. Leads V1, V2, and V5 are also possible candidates to detect CD based on their correlation with other leads.

D. Superclass disease lead correlation

We analyzed the correlation between each superclass of CVDs and the leads from the PTB-XL and combined dataset. In order to make a fair comparison between the datasets, we used the same groups for both. The results of our analysis are presented in the table below, which lists the leads most related to each subclass of CVDs based on their F1-scores.

TABLE IV: Lead Correlation for Each Superclass.

Superclass	Most relevant lead group	F1-score, PTB-XL	F1-score, CardioDiverse
ST-T Wave Abnormality	II, AVF, V3, V4, V5 ,V6	95.8	96
Hypertrophy	II, III, AVF, V1, V5, V6	92.7	93.2
Myocardial Infarction	II, AVF, AVL, V1, V2 ,\~	95.37	95.78
Conduction Disturbance	II, III, AVF, AVR, V1, V2	94.79	95.53

Table IV lists the five superclasses of the diseases (CD, HYP, STTC, and MI) and their most relevant leads based on the analysis of all possible combinations using PTB-XL and CardioDiverse. We can see that CardioDiverse results in better accuracy for all five classes. We observed a notable increase in the obtained results when compared to the previous case study focusing on CD. This improvement can be attributed to the inclusion of all five classes rather than just normal and CD. We selected the lead group associated with each superclass in each experiment segment to calculate the average F1 score. This strategic choice streamlined the classification process, facilitating the discrimination of each specific class from the others, not just the normal conditions.

VI. Conclusion

In this research, we built and evaluated CardioDiverse, an extensive ECG dataset that integrates a majority of the existing ECG datasets. Our analysis explored the relationship between leads and diseases using both the PTB-XL and CardioDiverse datasets. Findings indicate that CardioDiverse offers a more evenly distributed and inclusive resource for examining the relationship between leads and various diseases. Additionally, our examination of the association between each CVD superclass and the relevant leads in both datasets revealed that specific leads are more associated with certain CVD superclasses. Upon further analysis of notable subclasses, we determined that CardioDiverse encompasses a broader range of CVDs than any other existing dataset, establishing it as the benchmark for ECG studies. We've made CardioDiverse publicly available, aiming to catalyze further research into the early detection and diagnosis of cardiovascular diseases.

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